WRF hindcast report

Reproducible mesoscale hindcast and wind-resource screening

Parameter	Result
Simulation period	2025-01-10T00:00:00 to 2025-01-10T06:00:00 UTC
Domains	2
Primary analysis domain	d02
History files per domain	7
Report scope	Meteorology, surface fluxes, precipitation, PBL and 10 m wind-resource screening

Interpretation boundary

This technical report verifies model output and summarizes the simulated period. It is not a climatology, validation against observations, bankable wind-resource assessment, or substitute for physics sensitivity testing.

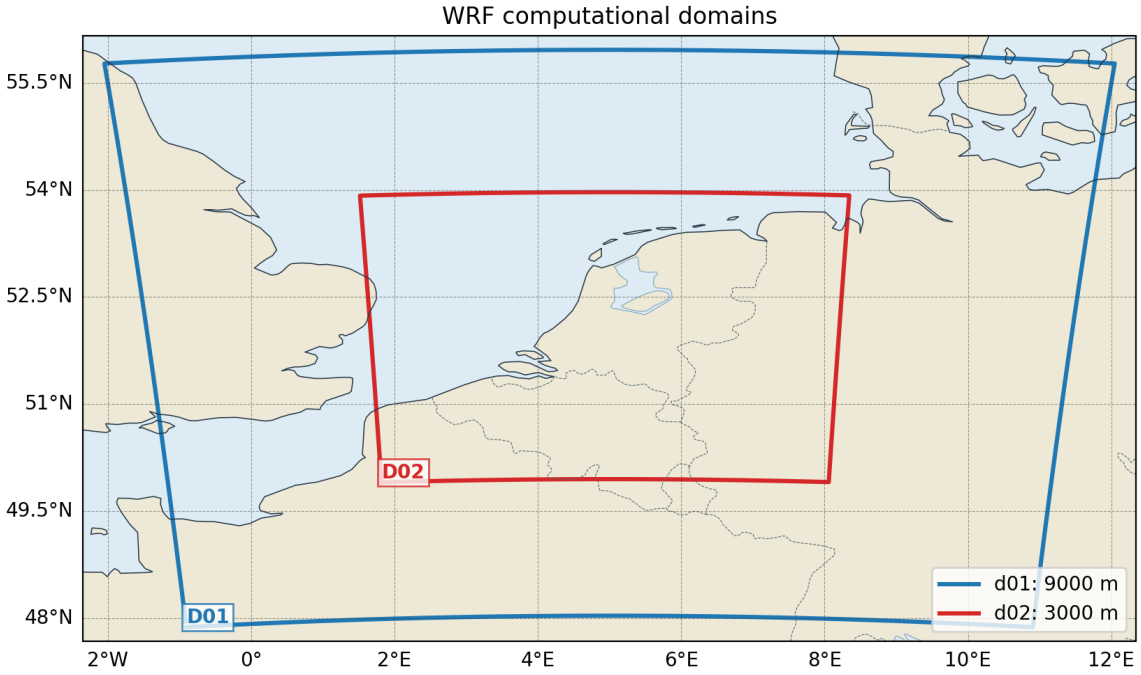


Figure 1. Computational-domain boundaries and nominal grid spacing.

Domain configuration comparison

Temporal configuration

Parameter	Result
Forcing interval	Not supplied; not inferred from history output
WRF output interval	60.0 minutes
Integration timestep	D01: 54 seconds, D02: 18 seconds
Restart checkpoint interval	None minutes
Resumed from	Fresh initialization

Grid

Parameter	D01	D02
Grid ID	1	2
Horizontal grid	99 x 99	150 x 150
Spacing	9000 x 9000 m	3000 x 3000 m
Vertical levels	39	39
Output times	7	7

Physics suite

Parameter	D01	D02
MP PHYSICS	4	4
RA LW PHYSICS	4	4
RA SW PHYSICS	4	4
SF SFCLAY PHYSICS	1	1
SF SURFACE PHYSICS	2	2
BL PBL PHYSICS	1	1
CU PHYSICS	1	1

Domain results comparison

Wind-resource screening

Parameter	D01	D02
Diagnostic height (m)	10.00	10.00
Mean wind speed (m s-1)	5.76	6.73
Maximum wind speed (m s-1)	13.09	13.09
50th-percentile wind speed (m s-1)	5.22	6.04
90th-percentile wind speed (m s-1)	10.38	11.43
95th-percentile wind speed (m s-1)	11.36	12.14
Mean direction FROM (degrees)	271.91	270.20
Mean air density (kg m-3)	1.27	1.28
Mean wind-power density (W m-2)	226.96	311.46

Parameter	D01	D02
Wind direction coordinates	earth-relative	earth-relative

Percentiles summarize sampled wind speed; they are not energy-yield exceedance P50/P90. Spatial means use unweighted grid cells.

Vertical-profile diagnostics

Parameter	D01	D02
Shear fit layer	40.0-200.0 m	40.0-200.0 m
Mean shear exponent	0.295	0.297

Parameter	D01	D02
Model TKE status	unavailable	unavailable
Model TKE source	not in output	not in output
Negative model TKE samples	not evaluated	not evaluated

D01 TKE unavailable: No supported QKE, TKE_PBL or TKE in this output. No resolved TKE inferred from hourly samples.

D02 TKE unavailable: No supported QKE, TKE_PBL or TKE in this output. No resolved TKE inferred from hourly samples.

Meteorological means and ranges

Parameter	D01	D02
T2	275 K (min 262; max 281)	275 K (min 266; max 280)
PSFC	1e+05 Pa (min 9.06e+04; max 1.02e+05)	1.01e+05 Pa (min 9.43e+04; max 1.02e+05)
Q2	0.00365 kg kg-1 (min 0.00154; max 0.00516)	0.00368 kg kg-1 (min 0.00217; max 0.00481)

Parameter	D01	D02
U10	4.11 m s-1 (min -6.19; max 11.5)	5.87 m s-1 (min -2.01; max 11.3)
V10	-1.01 m s-1 (min -11.1; max 6.48)	-1.01 m s-1 (min -9.19; max 5.22)
PBLH	414 m (min 0; max 1.28e+03)	443 m (min 0; max 1.4e+03)
RAINC	0.0195 mm (min 0; max 2.86)	0.0094 mm (min 0; max 2.2)
RAINNC	0.033 mm (min 0; max 4.06)	0.0288 mm (min 0; max 2.91)
HFX	0.994 W m-2 (min -77.5; max 98.5)	0.845 W m-2 (min -93.2; max 99.2)
LH	21.7 W m-2 (min -23.4; max 136)	26.8 W m-2 (min -24; max 136)
SWDOWN	0 W m-2 (min 0; max 0)	0 W m-2 (min 0; max 0)
GLW	210 W m-2 (min 0; max 333)	211 W m-2 (min 0; max 327)
UST	0.282 m s-1 (min 0.0001; max 1.12)	0.329 m s-1 (min 0.0001; max 1.03)

Precipitation coverage

Parameter	D01	D02
Period accumulation status	COMPLETE	COMPLETE
Valid grid-cell fraction	1.0	1.0
Cells with counter resets	0	0
Cells with missing samples	0	0

Precipitation period totals exclude cells with any counter reset or missing sample. Bucket/restart resets are not reconstructed; partial coverage is not a whole-domain estimate.

Quality-control overview

Parameter	D01	D02
QC status	ISSUES_FOUND	ISSUES_FOUND
Errors	0	0
Warnings	5	6

D01 QC details

Parameter	Result
WARNING - T2	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - PBLH	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - LH	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - GLW	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - UST	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1

D02 QC details

Parameter	Result
WARNING - T2	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - PBLH	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - RAINC	2 domain-mean jumps exceed 12 robust standard deviations; first at Time=5
WARNING - LH	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - GLW	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - UST	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1

Connected diagnostic groups

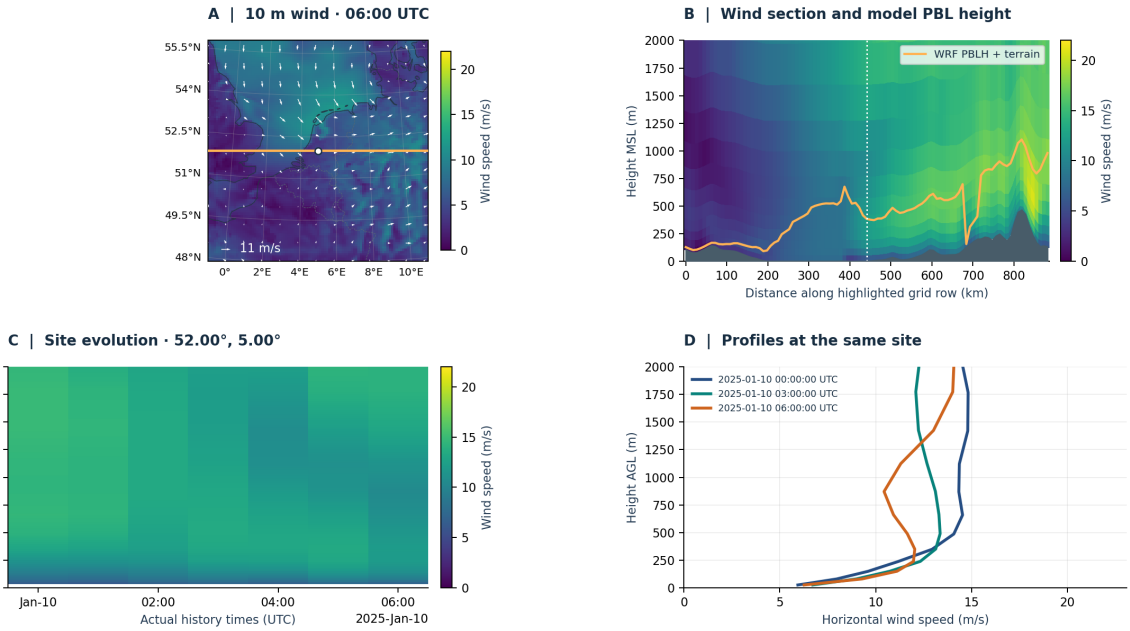
Parameter	D01	D02
Thermodynamics	COMPLETE	COMPLETE
Wind Resource	COMPLETE	COMPLETE
Flow Structure	COMPLETE	COMPLETE
Spatial	COMPLETE	COMPLETE
Spectral	COMPLETE	COMPLETE
Turbulence	UNAVAILABLE	UNAVAILABLE

D01 turbulence: Choose turbulence_average='time' and appropriate sampling/window explicitly; hourly weather variance is not silently labelled TKE

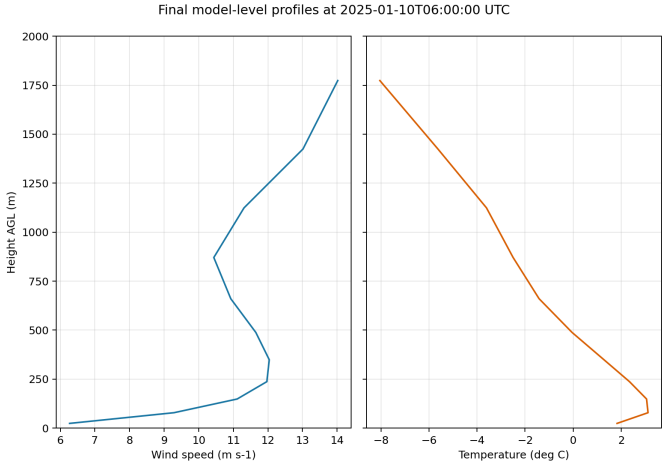
D02 turbulence: Choose turbulence_average='time' and appropriate sampling/window explicitly; hourly weather variance is not silently labelled TKE

Domain d01 figures

Grid ID 1 | 9000 m horizontal spacing

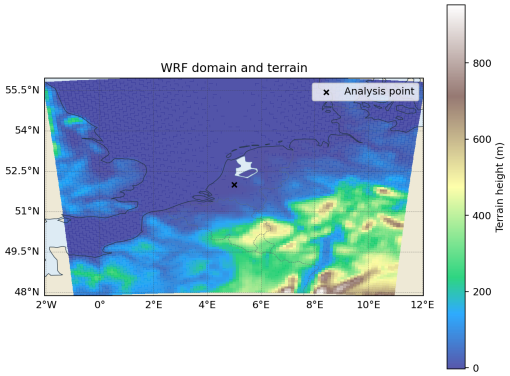


D01 wind overview. A: final 10 m wind; orange line marks the section and white marker the analysis site. B: horizontal wind speed on that native grid-row section, with model PBL height where available. C: site evolution on physical AGL heights. D: selected profiles. Colours share one scale. White regions have no supported data. Definitions and references appear in the methods appendix; PNG/SVG and a JSON processing sidecar accompany this report.

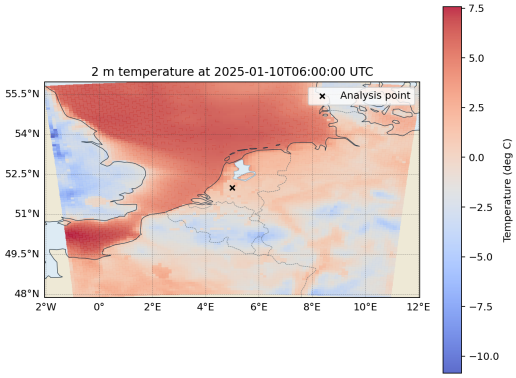


D01 additional temperature and turbulence profiles. See axis labels for the available diagnostics.

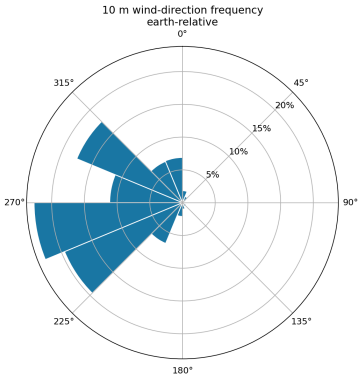
Domain d01 additional diagnostics



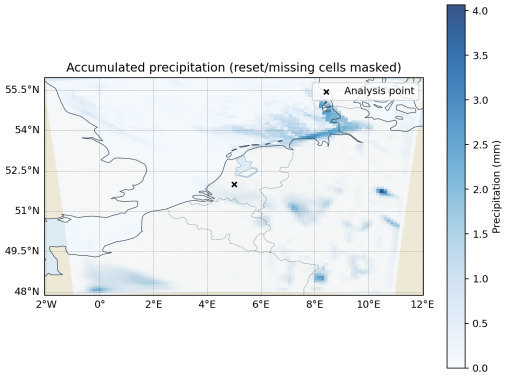
D01 Figure 1. Domain Terrain.



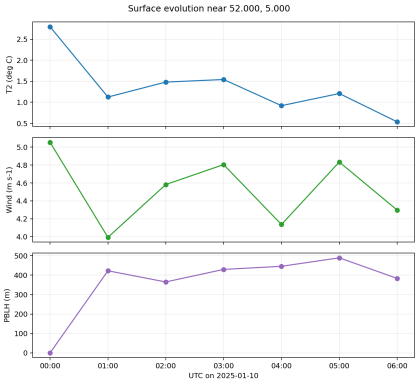
D01 Figure 2. Final Temperature.



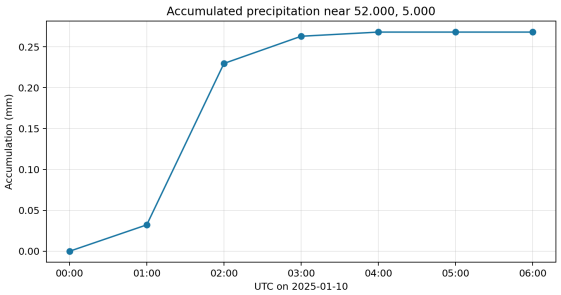
D01 Figure 3. Wind Rose.



D01 Figure 4. Accumulated Precipitation.

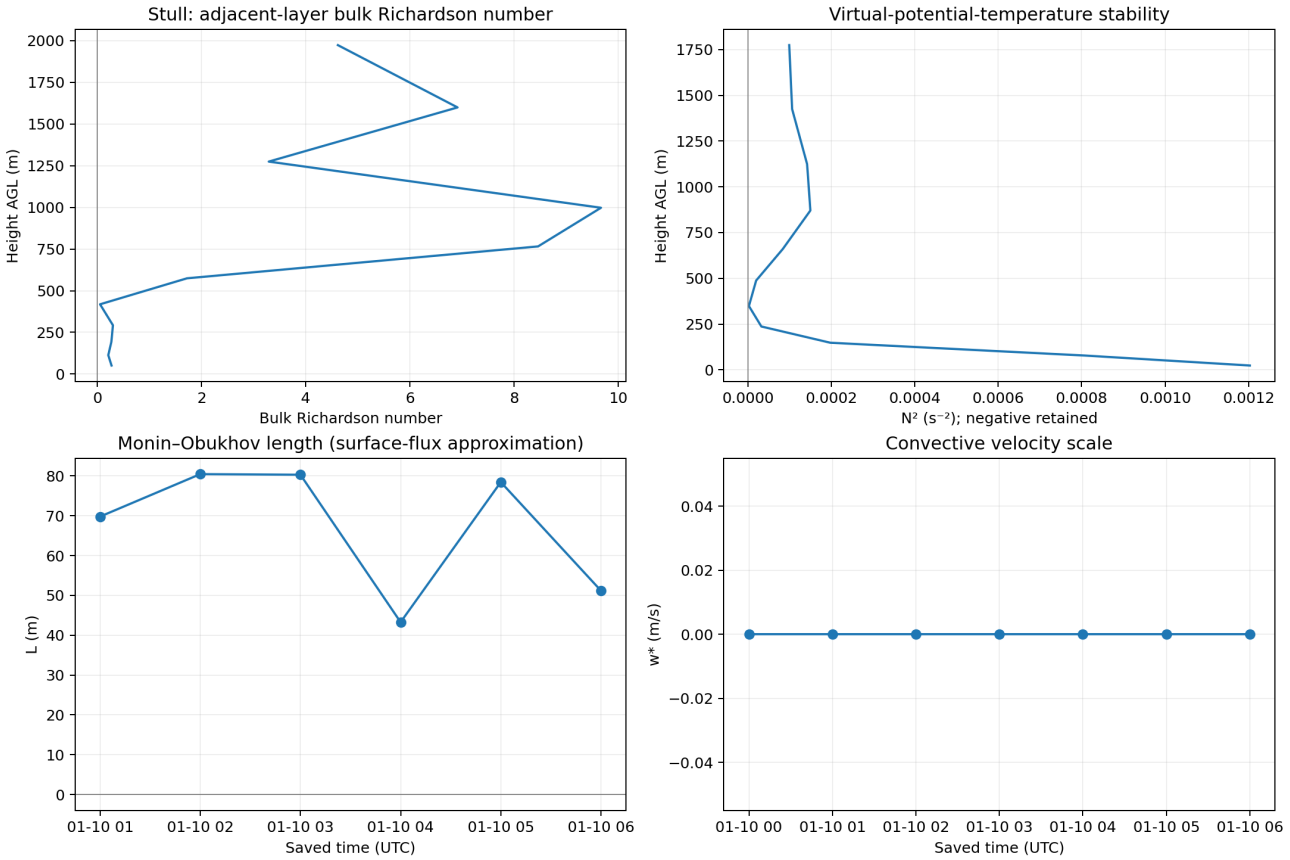


D01 Figure 5. Selected Point Timeseries.



D01 Figure 6. Precipitation Timeseries.

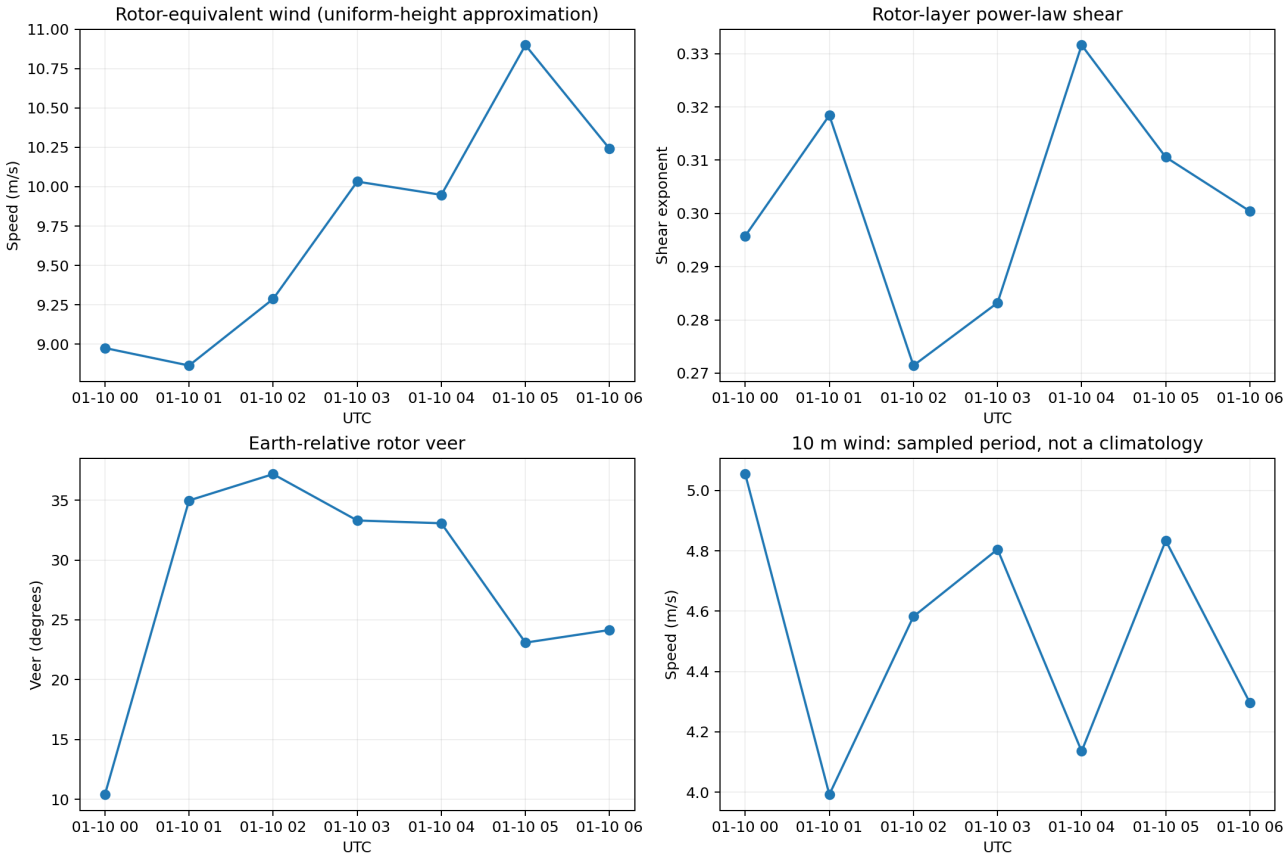
D01 | Thermodynamics



Stull bulk Ri and surface-flux similarity diagnostics. HFX/(rho cp) is converted to potential-temperature flux using theta/T before virtual-flux correction. Approximate virtual temperature neglects condensate. theta-e is a simplified screening formula; N² retains instability. Not a closed TKE budget.

Numerical arrays: d01_analysis_thermodynamics.npz. Companion PNG and SVG are saved separately.

D01 | Wind Resource

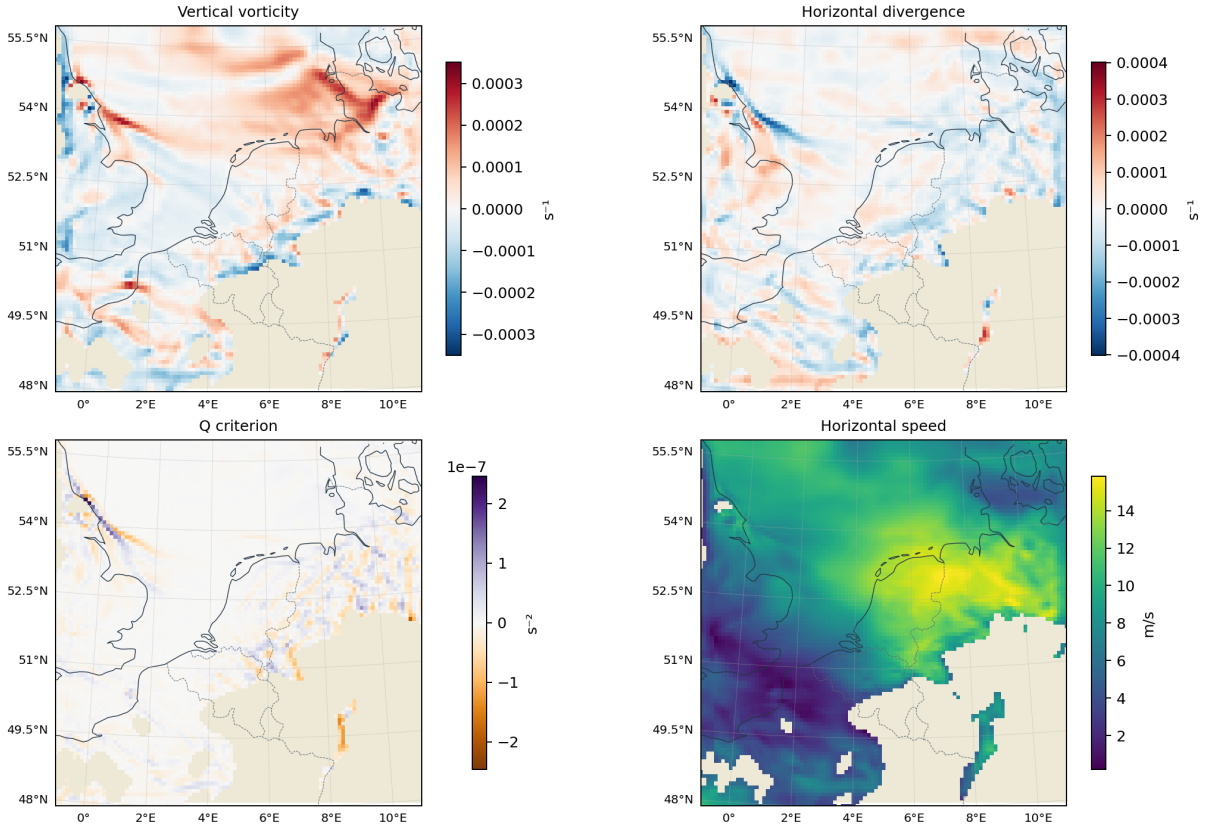


Research screening over 40–200 m AGL; incomplete columns excluded. Rotor speed uses uniform-height, not disk-area weights. Short-period Weibull estimates are not long-term resource estimates.

Numerical arrays: d01_analysis_wind_resource.npz. Companion PNG and SVG are saved separately.

D01 | Flow Structure

Final flow structure at 200 m MSL | Cartesian approximation

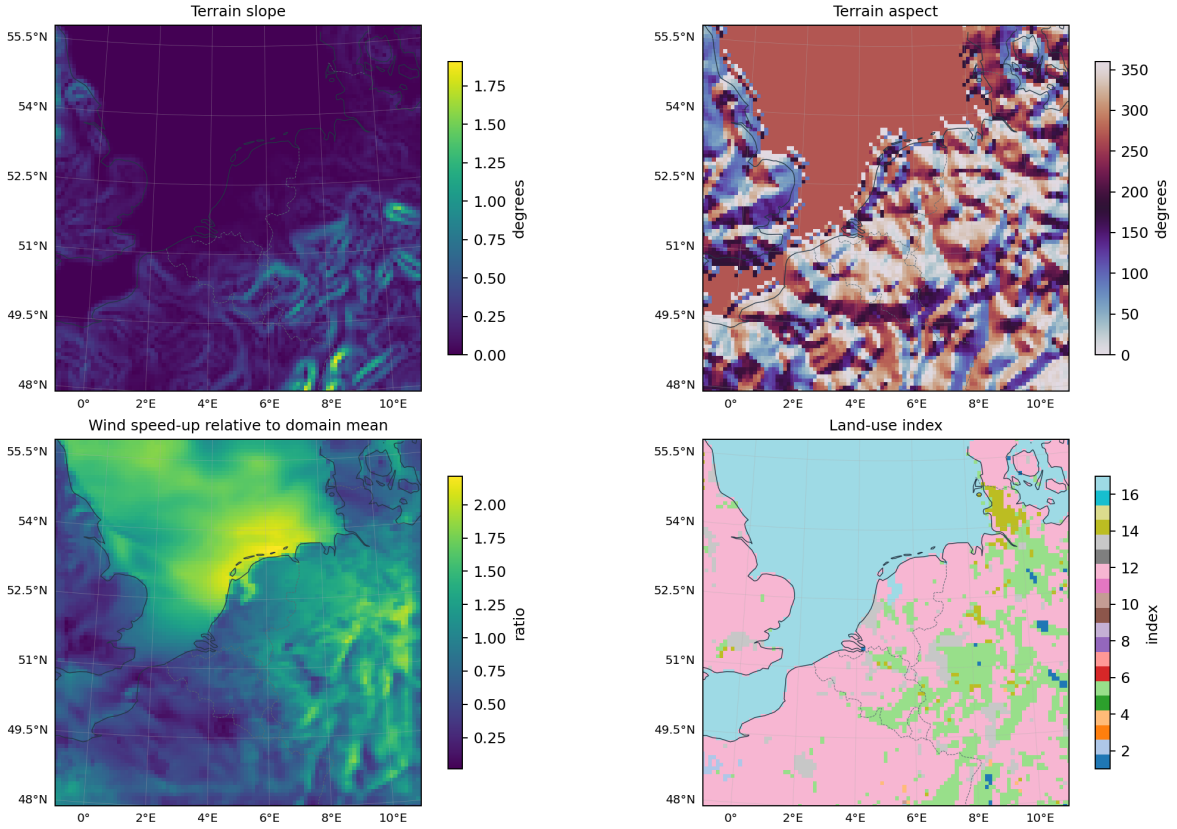


Wind interpolated to uniformly spaced MSL planes without extrapolation. Q/vorticity/divergence use nominal DX/DY Cartesian derivatives; projection metrics and varying grid basis are neglected. This approximation is explicit, not an exact WRF metric-aware operator.

Numerical arrays: d01_analysis_flow_structure.npz. Companion PNG and SVG are saved separately.

D01 | Spatial

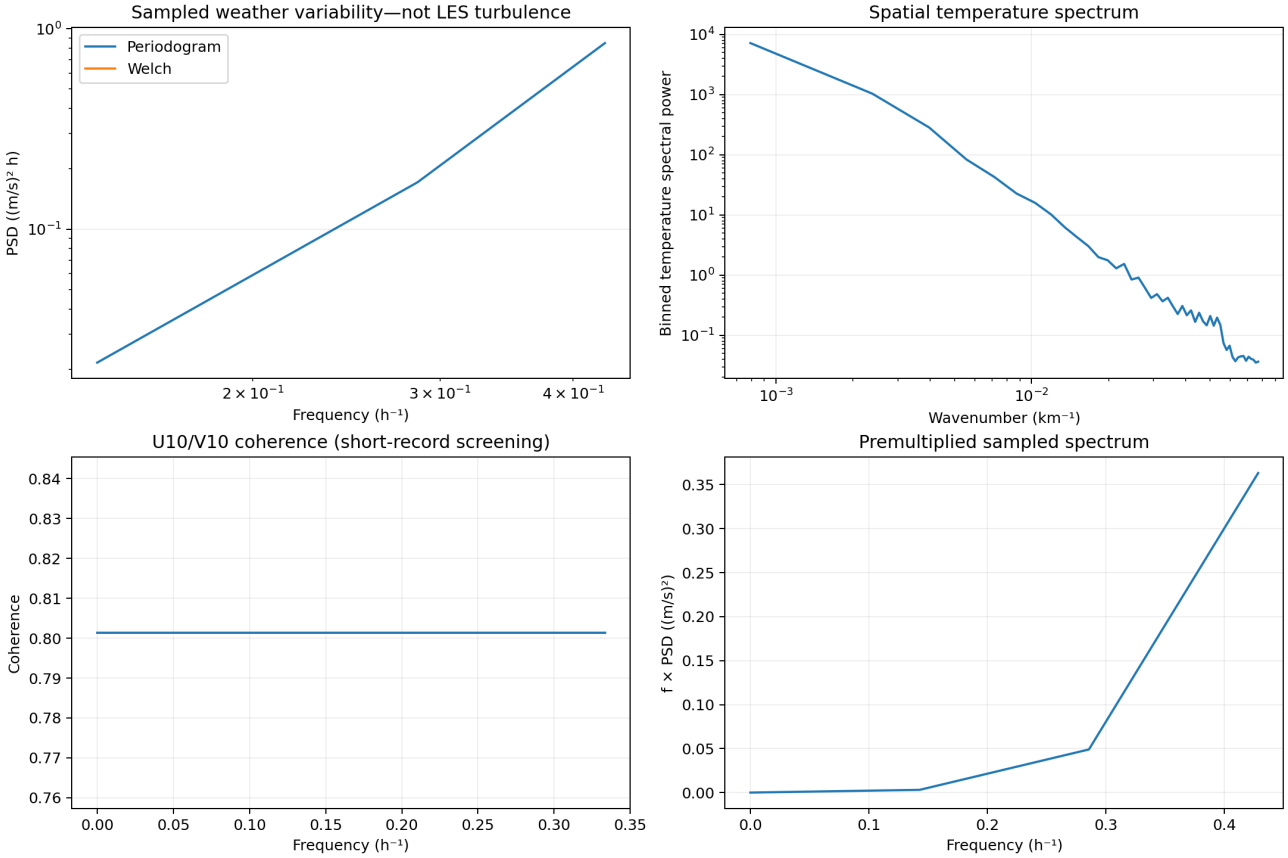
Terrain, land-use and wind exposure



Nominal-grid terrain derivatives. Speed-up uses the instantaneous unweighted domain-mean speed as a reference, not a paired control simulation. Land-use numbers are categorical codes.

Numerical arrays: d01_analysis_spatial.npz. Companion PNG and SVG are saved separately.

D01 | Spectral

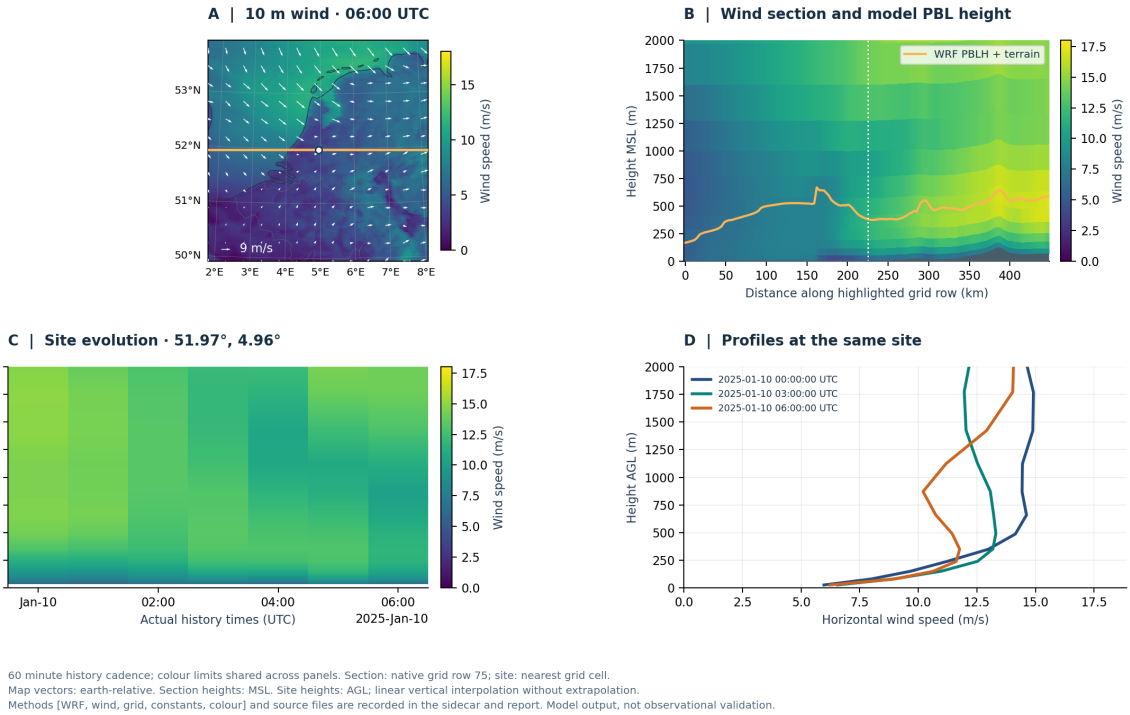


Temporal spectra describe the selected sampling/period only. No -5/3 turbulent-range claim is made. FFT spatial filters assume periodic boundaries. Coherence from few samples is poorly constrained. Coarsened arrays and conservation diagnostics are saved with the spectra.

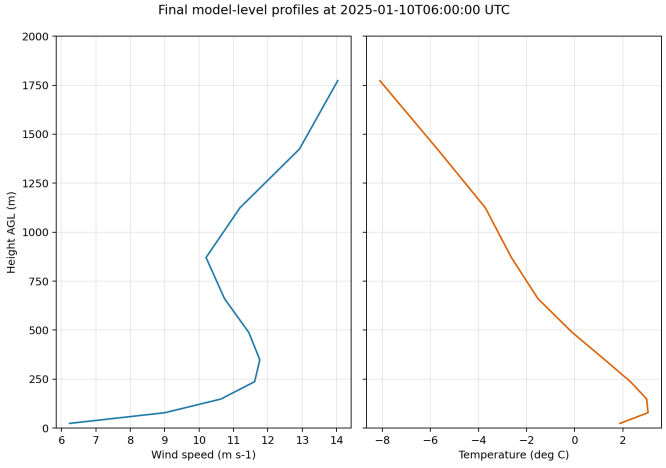
Numerical arrays: d01_analysis_spectral.npz. Companion PNG and SVG are saved separately.

Domain d02 figures

Grid ID 2 | 3000 m horizontal spacing

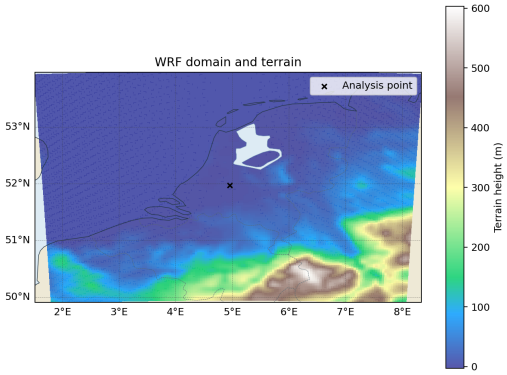


D02 wind overview. A: final 10 m wind; orange line marks the section and white marker the analysis site. B: horizontal wind speed on that native grid-row section, with model PBL height where available. C: site evolution on physical AGL heights. D: selected profiles. Colours share one scale. White regions have no supported data. Definitions and references appear in the methods appendix; PNG/SVG and a JSON processing sidecar accompany this report.

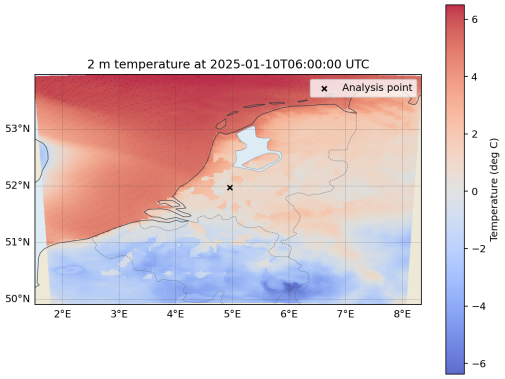


D02 additional temperature and turbulence profiles. See axis labels for the available diagnostics.

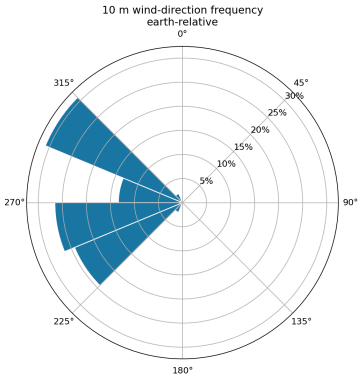
Domain d02 additional diagnostics



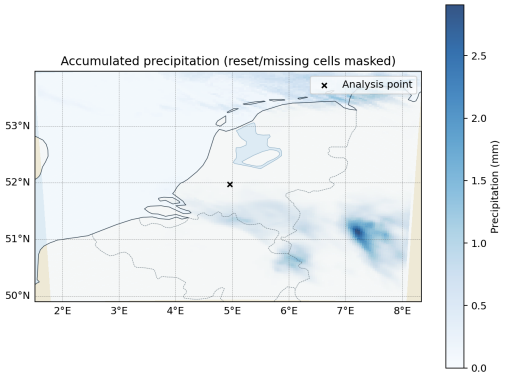
D02 Figure 1. Domain Terrain.



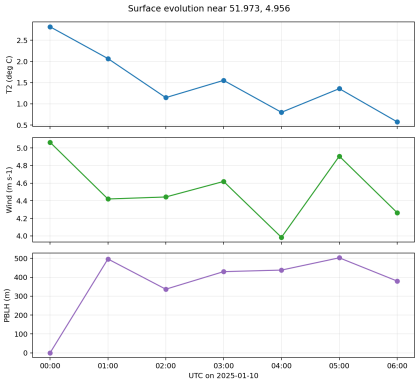
D02 Figure 2. Final Temperature.



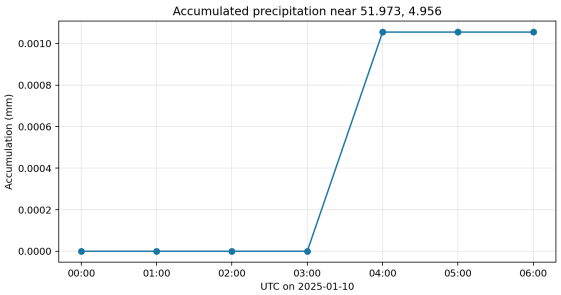
D02 Figure 3. Wind Rose.



D02 Figure 4. Accumulated Precipitation.

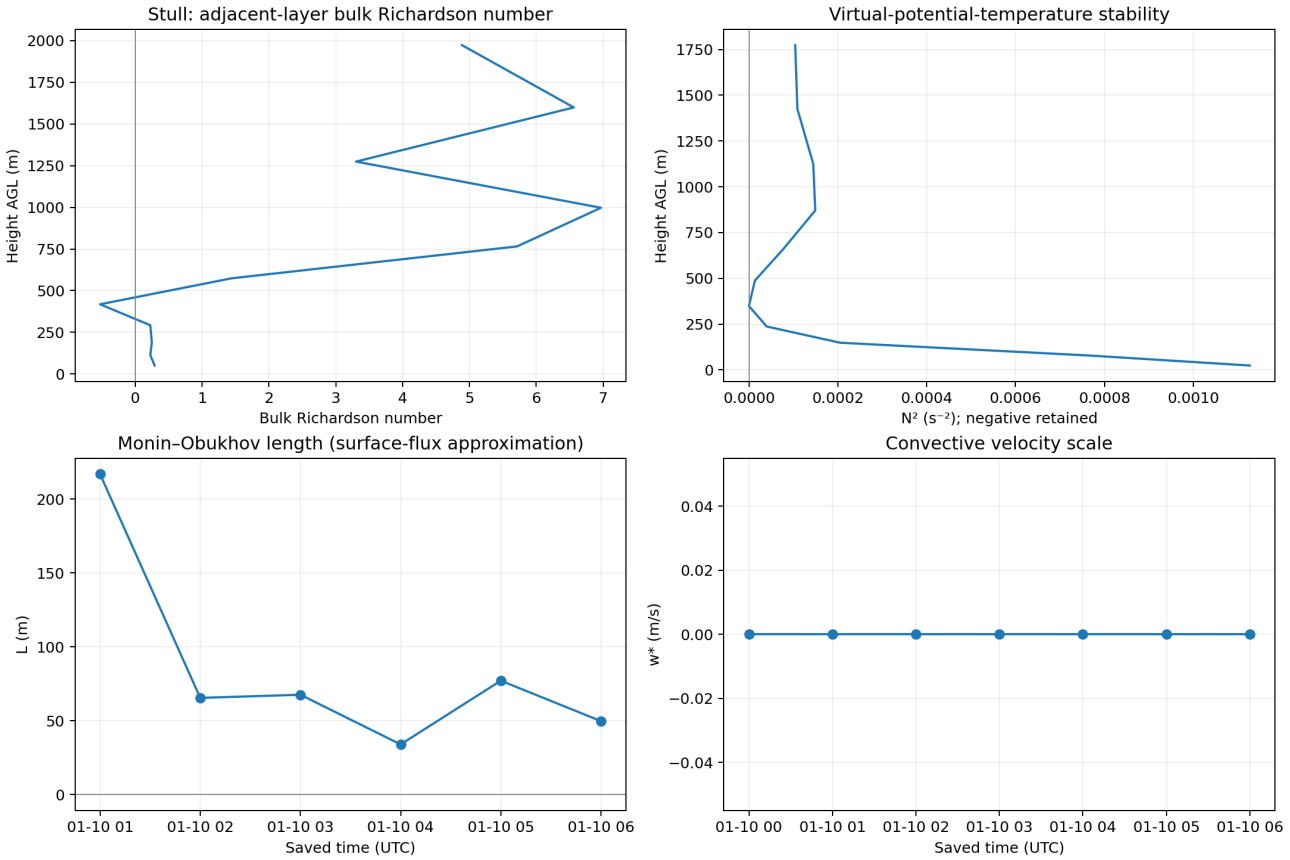


D02 Figure 5. Selected Point Timeseries.



D02 Figure 6. Precipitation Timeseries.

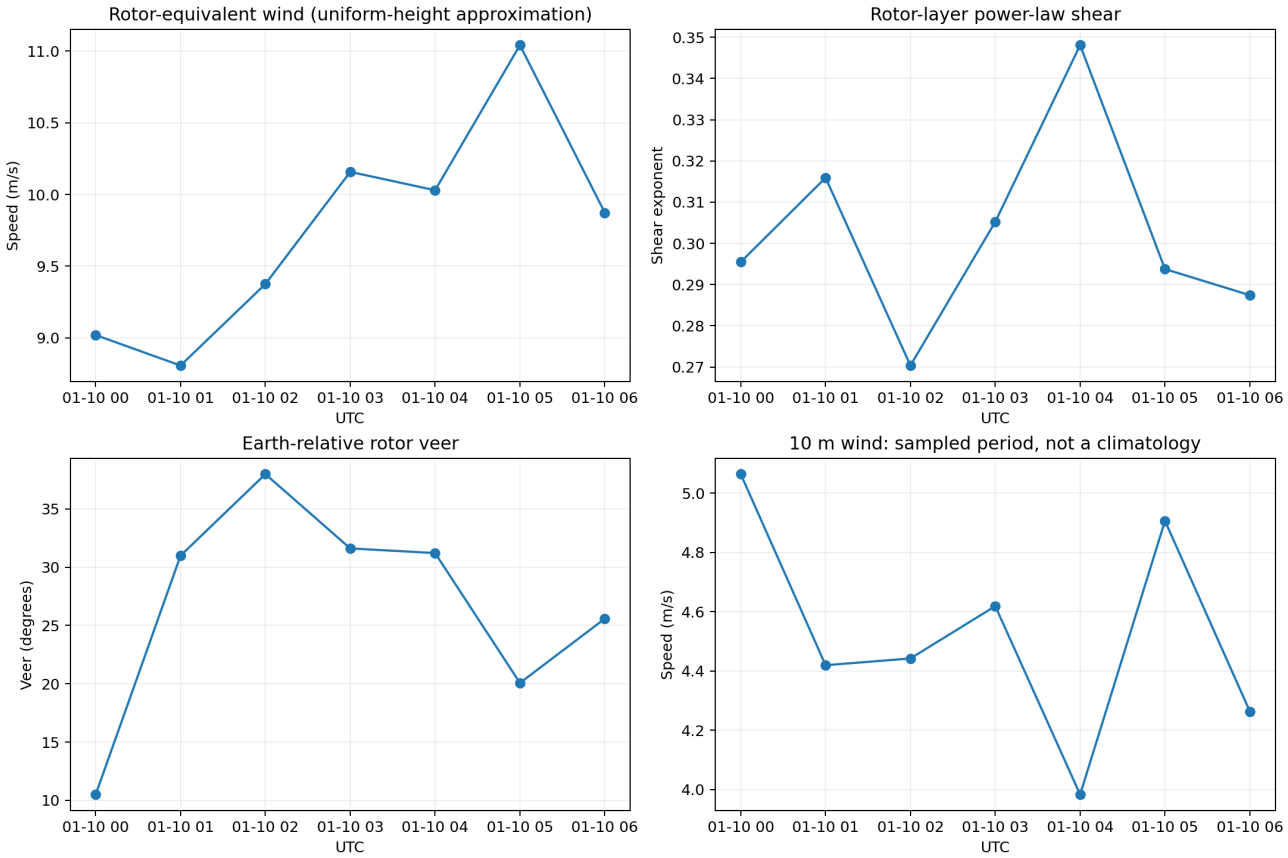
D02 | Thermodynamics



Stull bulk Ri and surface-flux similarity diagnostics. HFX/(rho cp) is converted to potential-temperature flux using theta/T before virtual-flux correction. Approximate virtual temperature neglects condensate. theta-e is a simplified screening formula; N² retains instability. Not a closed TKE budget.

Numerical arrays: d02_analysis_thermodynamics.npz. Companion PNG and SVG are saved separately.

D02 | Wind Resource

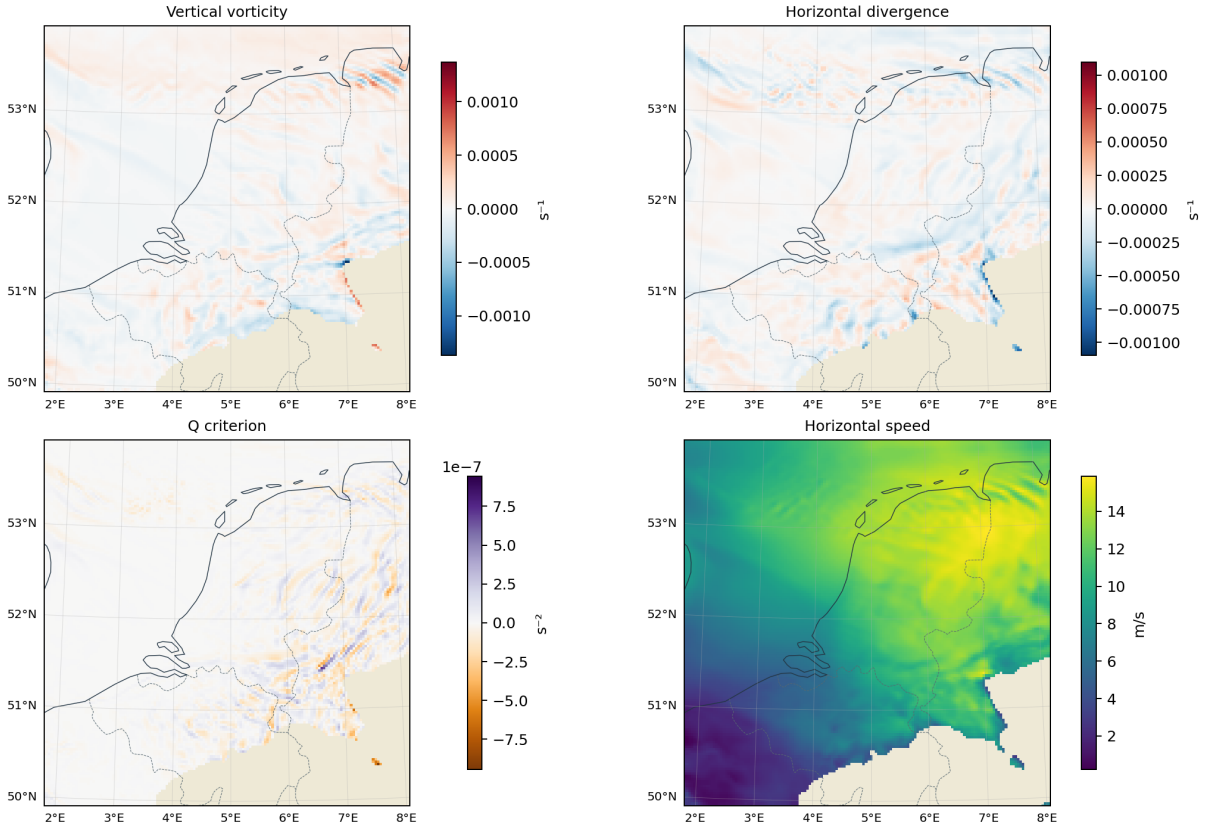


Research screening over 40–200 m AGL; incomplete columns excluded. Rotor speed uses uniform-height, not disk-area weights. Short-period Weibull estimates are not long-term resource estimates.

Numerical arrays: d02_analysis_wind_resource.npz. Companion PNG and SVG are saved separately.

D02 | Flow Structure

Final flow structure at 200 m MSL | Cartesian approximation

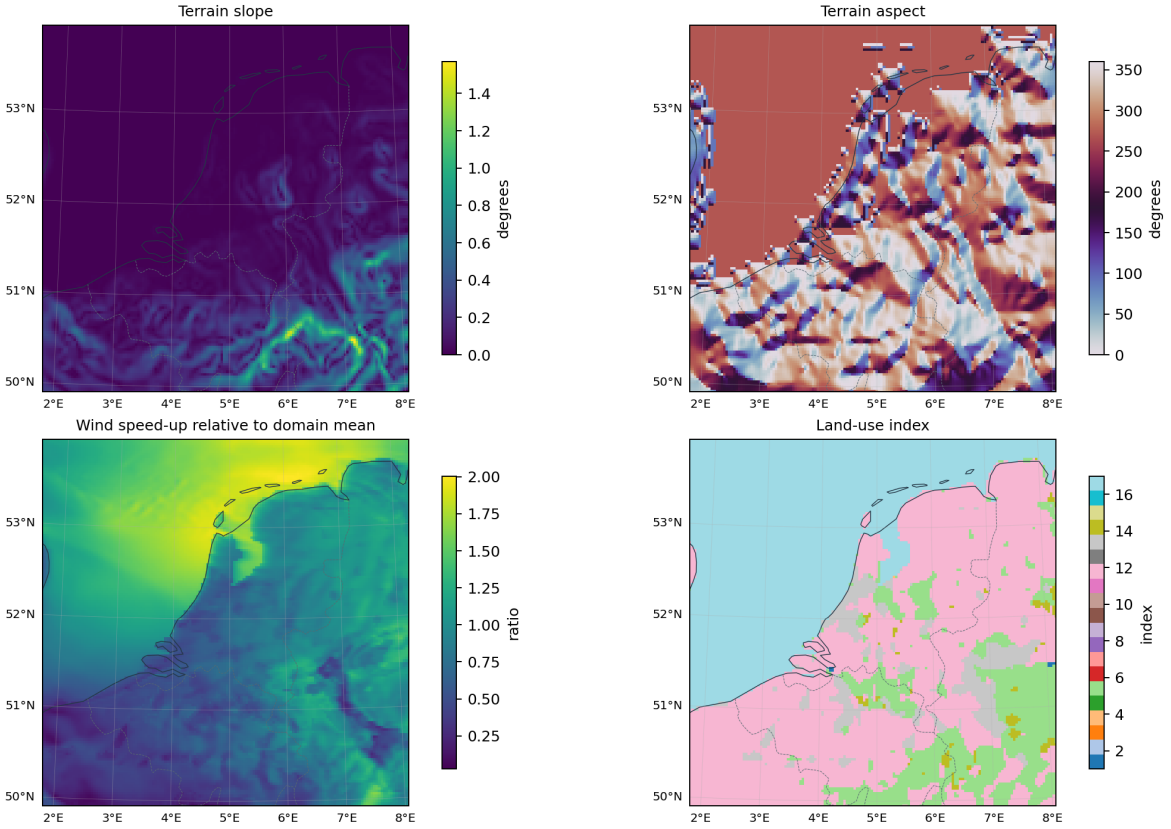


Wind interpolated to uniformly spaced MSL planes without extrapolation. Q/vorticity/divergence use nominal DX/DY Cartesian derivatives; projection metrics and varying grid basis are neglected. This approximation is explicit, not an exact WRF metric-aware operator.

Numerical arrays: d02_analysis_flow_structure.npz. Companion PNG and SVG are saved separately.

D02 | Spatial

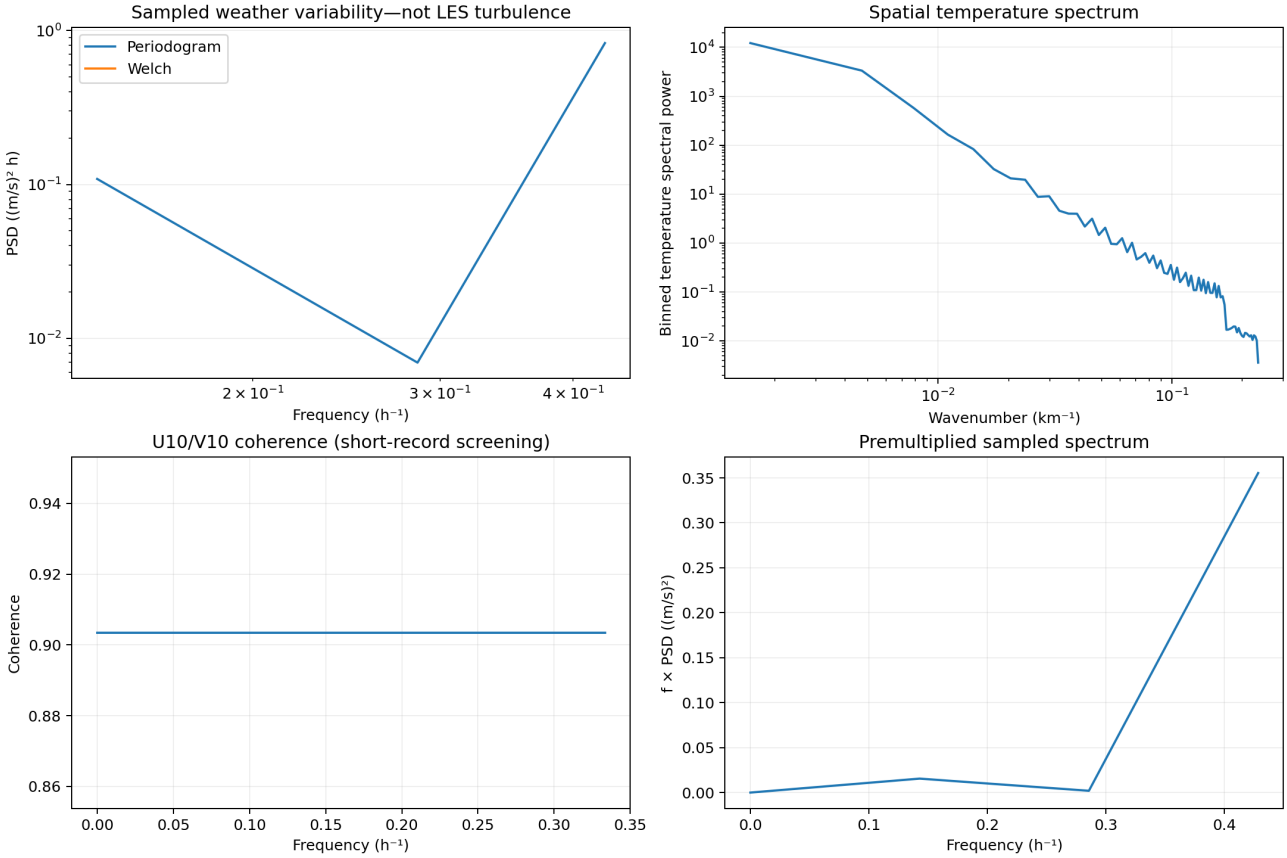
Terrain, land-use and wind exposure



Nominal-grid terrain derivatives. Speed-up uses the instantaneous unweighted domain-mean speed as a reference, not a paired control simulation. Land-use numbers are categorical codes.

Numerical arrays: d02_analysis_spatial.npz. Companion PNG and SVG are saved separately.

D02 | Spectral



Temporal spectra describe the selected sampling/period only. No -5/3 turbulent-range claim is made. FFT spatial filters assume periodic boundaries. Coherence from few samples is poorly constrained. Coarsened arrays and conservation diagnostics are saved with the spectra.

Numerical arrays: d02_analysis_spectral.npz. Companion PNG and SVG are saved separately.

Scientific methods and references

Methods describe output processing. Numerical checks do not establish atmospheric accuracy; no observation-based validation is performed.

Wind Direction

Meteorological direction FROM; circular mean, excluding calm samples (<1e-6 m/s). Earth-relative when rotation fields exist, otherwise explicitly grid-relative.

Wind Percentiles

Sample wind-speed percentiles; not energy-yield P50/P90 exceedance estimates.

Summary Averaging

Unweighted grid-cell and history-sample statistics, not an area-weighted or time-weighted climatology.

Temperature

(T+300)*((P+PB)/100000)**(2/7) - 273.15 deg C; WRF dry Rd=287, cp=7*Rd/2, theta0=300 K. The perturbation dry-theta convention must match the supplied WRF output.

Screening Air Density

PSFC/(287.05*T2), when available; otherwise 1.225 kg/m3. This is an approximate dry-air 10 m screening proxy, not a moist model-level density diagnostic.

Tke

Model-written QKE/2 (MYNN), TKE_PBL directly, or dynamics TKE directly, selected individually, never summed. Interface values are averaged to mass levels when needed. Negative samples are retained. Not resolved/total TKE or proof of spurious energy production. A closed budget requires saved production, dissipation and transport tendencies.

Shear

Least-squares log(speed) versus log(height AGL), using positive speeds within the configured layer; a diagnostic power-law fit.

Precipitation

Counter differences of RAINC+RAINNC. Any reset or nonfinite sample masks that cell's period total; valid-cell fraction and affected counts are reported. Bucket/restart reconstruction is not supplied. Affected site time series retain the raw counter change with a warning.

References

[wrf] Skamarock et al. (2019), A Description of the Advanced Research WRF Model Version 4. https://doi.org/10.5065/1dfh-6p97

[wind] WRF-Python: earth-relative wind components (uvmet). https://wrf-python.readthedocs.io/en/latest/user_api/generated/wrf.uvmet.html

[grid] WRF-Python: diagnostic and height conventions. https://wrf-python.readthedocs.io/en/latest/user_api/generated/wrf.getvar.html

[constants] WRF v4.7.1 module_model_constants.F: gravity, reference temperature and dry-air constants. https://github.com/wrf-model/WRF/blob/v4.7.1/share/module_model_constants.F

[colour] Crameri, Shephard and Heron (2020), The misuse of colour in science communication. https://doi.org/10.1038/s41467-020-19160-7

[stull] Stull (1988), An Introduction to Boundary Layer Meteorology: Richardson number and partial TKE budget. https://doi.org/10.1007/978-94-009-3027-8

[q_criterion] Hunt, Wray and Moin (1988), Eddies, streams, and convergence zones in turbulent flows. https://ntrs.nasa.gov/citations/19890015184

[welch] Welch (1967), Power spectra from modified periodograms. https://doi.org/10.1109/TAU.1967.1161901

[pope] Pope (2000), Turbulent Flows: resolved Reynolds covariances. https://doi.org/10.1017/CBO9780511840531

[rotor] Wagner et al. (2014), Rotor-equivalent wind speed; this package uses a uniform-height approximation. https://doi.org/10.1088/1742-6596/524/1/012108

[weibull] Justus et al. (1978), Methods for estimating wind speed frequency distributions. https://doi.org/10.1175/1520-0450(1978)017<0350:MFEWSF>2.0.CO;2

Reproducibility and provenance

Parameter	Result
inputs	["wrfout_d01_2025-01-10_00:00:00", "wrfout_d02_2025-01-10_00:00:00", " wrfout_d01_2025-01-10_01:00:00", "wrfout_d02_2025-01-10_01:00:00", " wrfout_d01_2025-01-10_02:00:00", "wrfout_d02_2025-01-10_02:00:00", " wrfout_d01_2025-01-10_03:00:00", "wrfout_d02_2025-01-10_03:00:00", " wrfout_d01_2025-01-10_04:00:00", "wrfout_d02_2025-01-10_04:00:00", " wrfout_d01_2025-01-10_05:00:00", "wrfout_d02_2025-01-10_05:00:00", " wrfout_d01_2025-01-10_06:00:00", "wrfout_d02_2025-01-10_06:00:00"]
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